Angular Demos

ng new, ng serve,
(ng-)bootstrap library
basic elements: button, menu, tab, ..

arnaud.nauwynck@gmail.com

Course Esilv 2025

This document:
https://github.com/Arnaud-Nauwynck/presentations/web/
angular-demos-1-ng-new-ngbootstrap.pdf

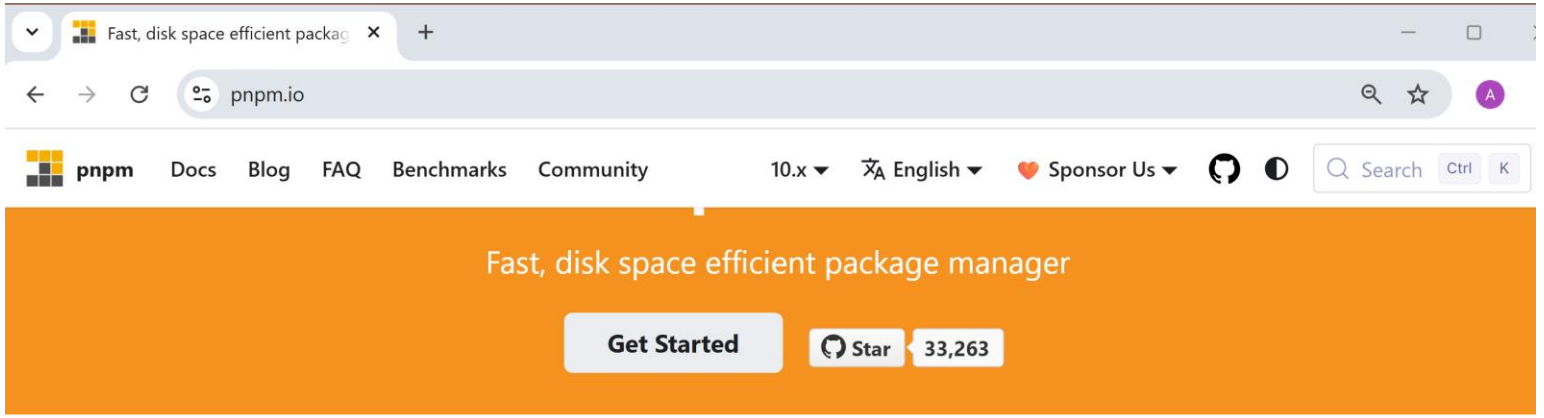
2 pre-requisites:

1/ install **node.js**
2/ **$ npm install -g @angular/cli**

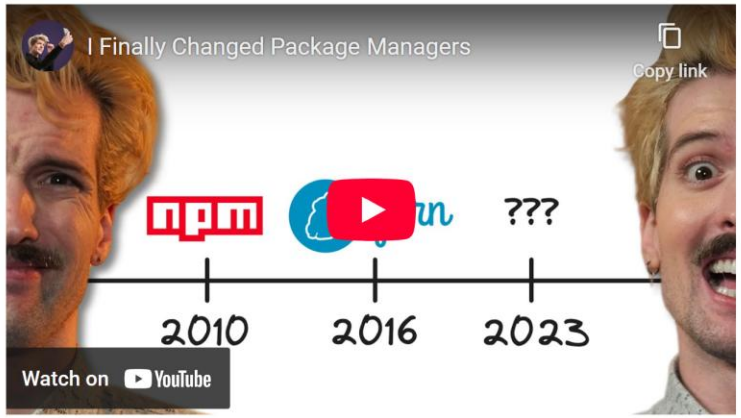
Also recommended :
3/ have 1G disk free on your PC
4/ install Intellij or Visual Studio Code
5/ network bandwidth
6/ patience

(Optional) Using pnpm instead of npm …

https://pnpm.io



npm install -g pnpm@latest-10



Fast

pnpm is up to 2x faster than npm

Efficient

Files inside node_modules are cloned or hard linked from a single content-addressable storage

Supports monorepos

pnpm has built-in support for multiple packages in a repository

Strict

pnpm creates a non-flat node_modules by default, so code has no access to arbitrary packages

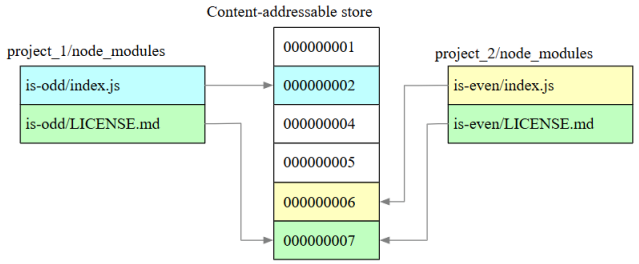
Pnpm Motivations

Introduction › Motivation

Version: 10.x

Motivation

Saving disk space



When using npm, if you have 100 projects using a dependency, you will have 100 copies of that dependency saved on disk. With pnpm, the dependency will be stored in a content-addressable store, so:

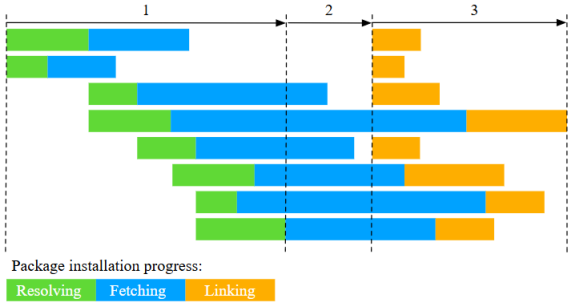
1. If you depend on different versions of the dependency, only the files that differ are added to the store. For instance, if it has 100 files, and a new version has a change in only one of those files, `pnpm update` will only add 1 new file to the store, instead of cloning the entire dependency just for the singular change.
2. All the files are saved in a single place on the disk. When packages are installed, their files are hard-linked from that single place, consuming no additional disk space. This allows you to share dependencies of the same version across projects.

As a result, you save a lot of space on your disk proportional to the number of projects and dependencies, and you have a lot faster installations!

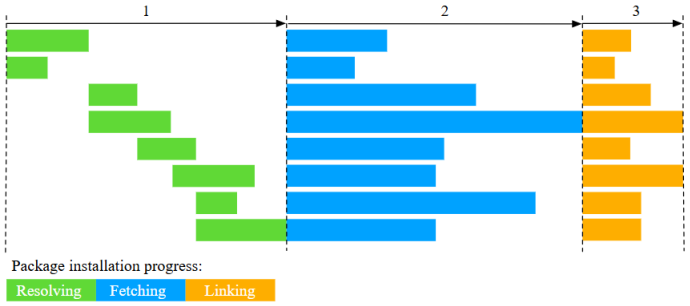
Boosting installation speed

pnpm performs installation in three stages:

1. Dependency resolution. All required dependencies are identified and fetched to the store.
2. Directory structure calculation. The `node_modules` directory structure is calculated based on the dependencies.
3. Linking dependencies. All remaining dependencies are fetched and hard linked from the store to `node_modules`.



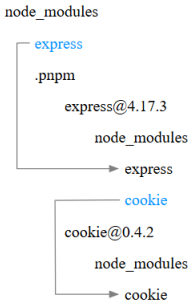
This approach is significantly faster than the traditional three-stage installation process of resolving, fetching, and writing all dependencies to `node_modules`.



Creating a non-flat node_modules directory

When installing dependencies with npm or Yarn Classic, all packages are hoisted to the root of source code has access to dependencies that are not added as dependencies to the project.

By default, pnpm uses symlinks to add only the direct dependencies of the project into the roo



SMALLER (disk space)

FASTER

BETTER

(Optional) Configuring « ng cli » to use pnpm instead of npm

```
ng config -g cli.packageManager pnpm
```

Notice: neither package.json nor angular.jso files need to be changed to use pnpm.

But all commands « npm install » need to be replaced by « pnpm install »
and « ng new » is calling « ${cli.packageManager} install »

$ ng new

Create a new application

Upgrade existing @angular/cli .. If needed

```
$ ng update @angular/cli
```

or

```
$ npm uninstall @angular/cli
$ npm install –g @angular/cli
```

Upgrade existing angular project … if needed

$ ng update

```
$ ng update
Using package manager: npm
Collecting installed dependencies...
Found 22 dependencies.
    We analyzed your package.json and everything seems to be in order. Good work!
```

« ng new » alternatives

1/ $ **git clone** https://github.com/${org}/${repo}

2/ **jhispter**

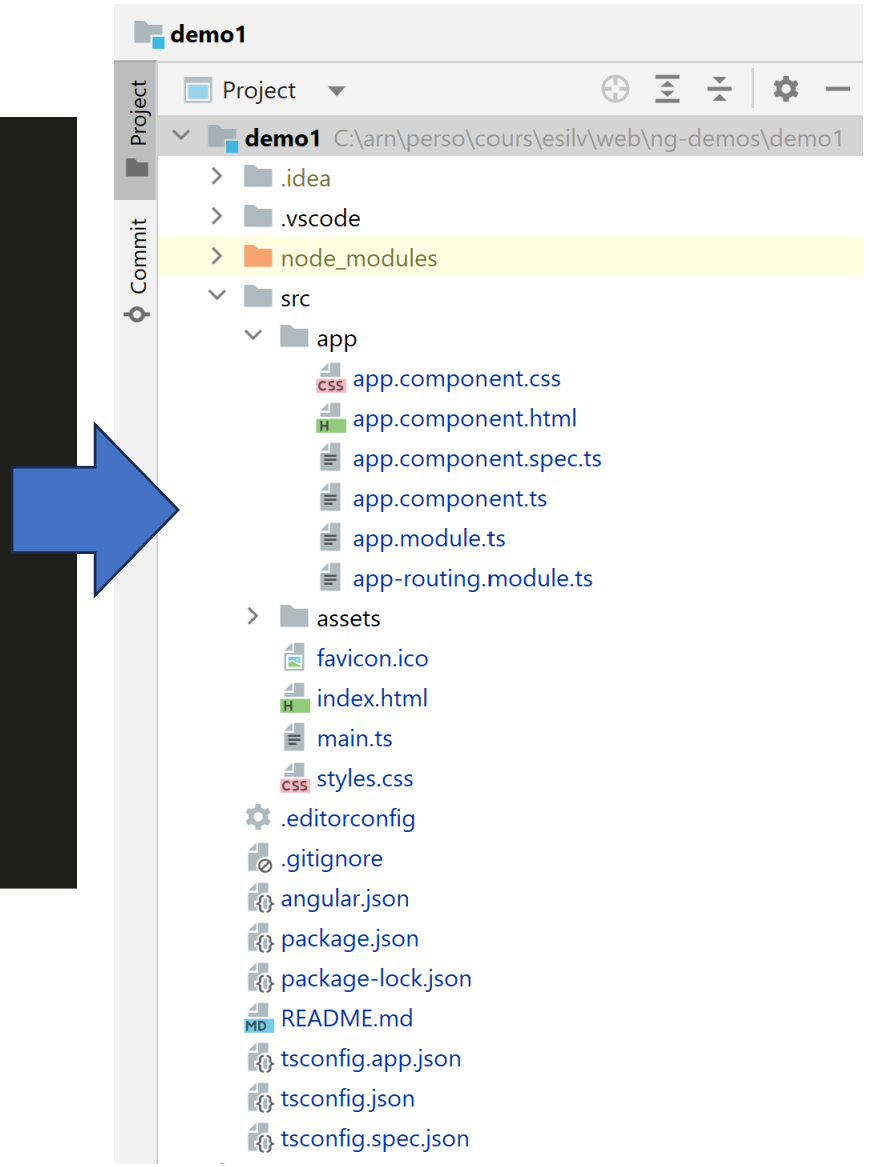
3/ read docs, edit package.json + angular.json
… NOT recommended

Jhipster = Java springboot(backend) + angular(frontend) app generator



ng new

```
$ ng new
? What name would you like to use for the new workspace and initial project? demo1
? Would you like to add Angular routing? Yes
? Which stylesheet format would you like to use? CSS
CREATE demo1/angular.json (2695 bytes)
CREATE demo1/package.json (1036 bytes)
CREATE demo1/README.md (1059 bytes)
CREATE demo1/tsconfig.json (901 bytes)
CREATE demo1/.editorconfig (274 bytes)
CREATE demo1/.gitignore (548 bytes)
CREATE demo1/tsconfig.app.json (263 bytes)
CREATE demo1/tsconfig.spec.json (273 bytes)
CREATE demo1/.vscode/extensions.json (130 bytes)
CREATE demo1/.vscode/launch.json (474 bytes)
CREATE demo1/.vscode/tasks.json (938 bytes)
CREATE demo1/src/favicon.ico (948 bytes)
CREATE demo1/src/index.html (291 bytes)
CREATE demo1/src/main.ts (214 bytes)
CREATE demo1/src/styles.css (80 bytes)
```



$ ng serve => http://localhost:4200

```
$ ng serve
√ Browser application bundle generation complete.

Initial Chunk Files   | Names         |  Raw Size
vendor.js             | vendor        |   2.04 MB |
polyfills.js          | polyfills     | 314.82 kB |
styles.css, styles.js | styles        | 209.94 kB |
main.js               | main          |  48.08 kB |
runtime.js            | runtime       |   6.51 kB |

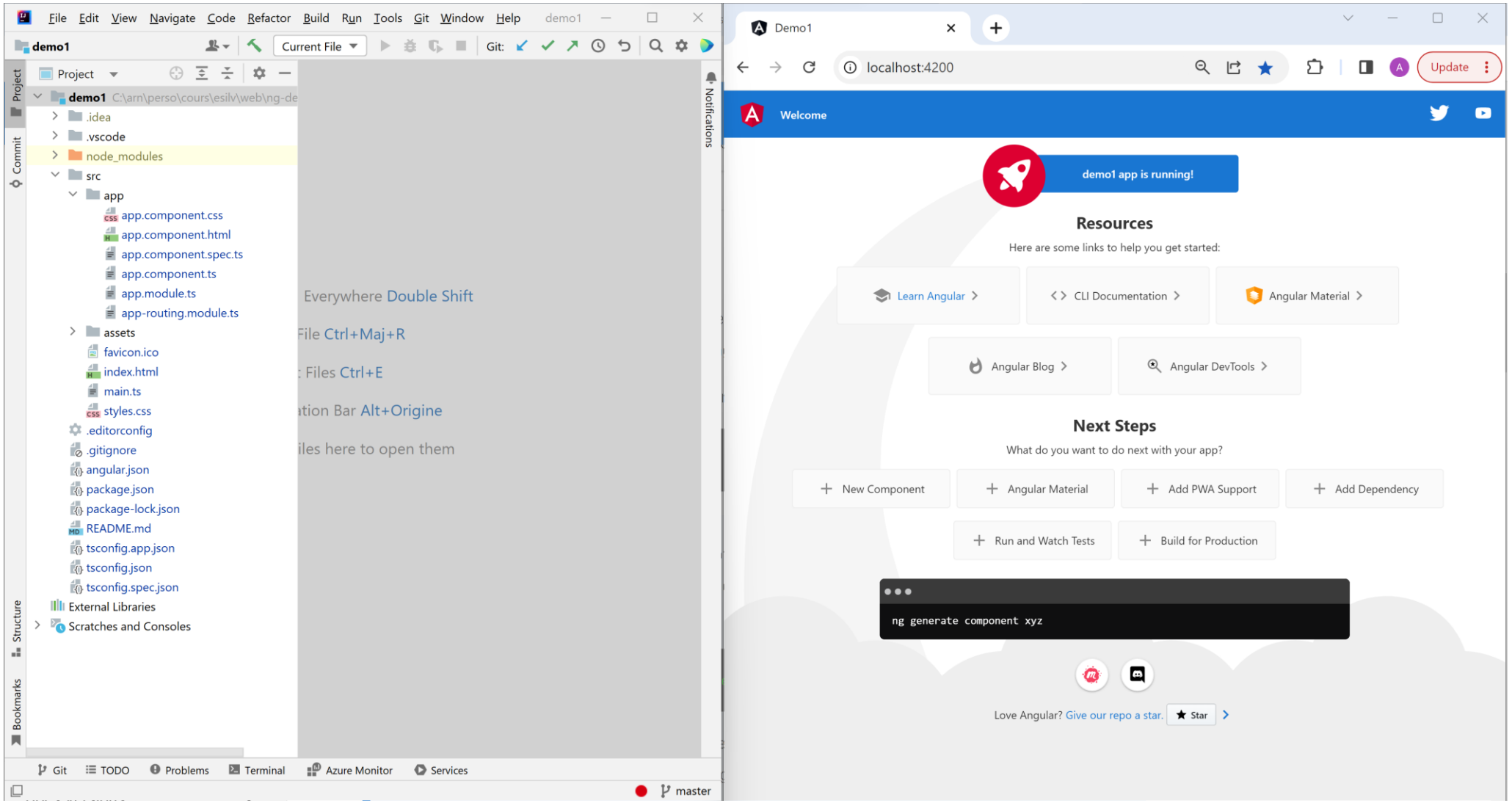
                      | Initial Total |   2.61 MB

Build at: 2023-07-09T10:06:15.879Z - Hash: 70e19beef187b407 - Time: 22482ms

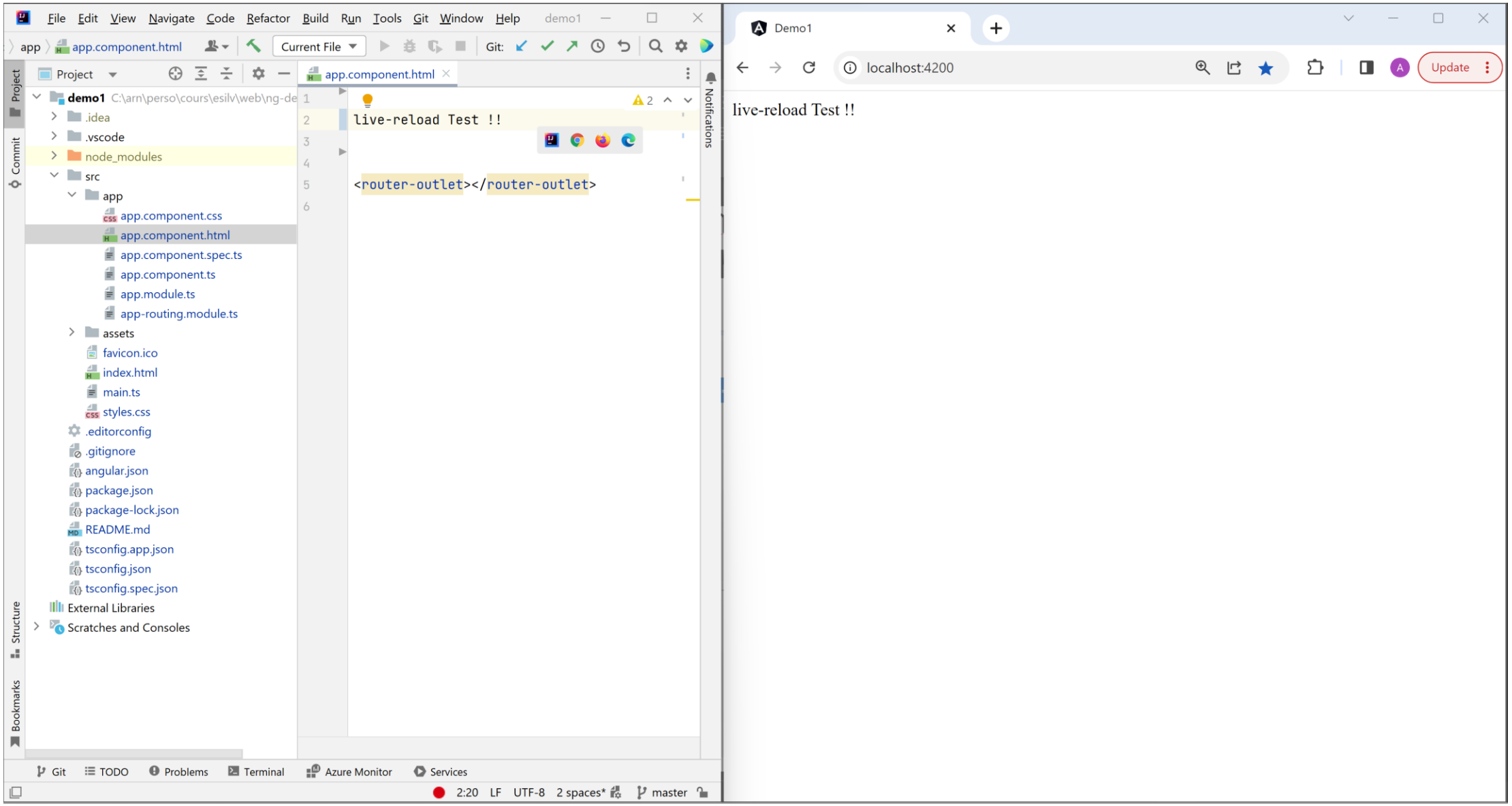
** Angular Live Development Server is listening on localhost:4200, open your browser on http://localhost:4200/ **


√ Compiled successfully.
```

Demo
IntelliJ (left) + Chrome (right) + ng serve

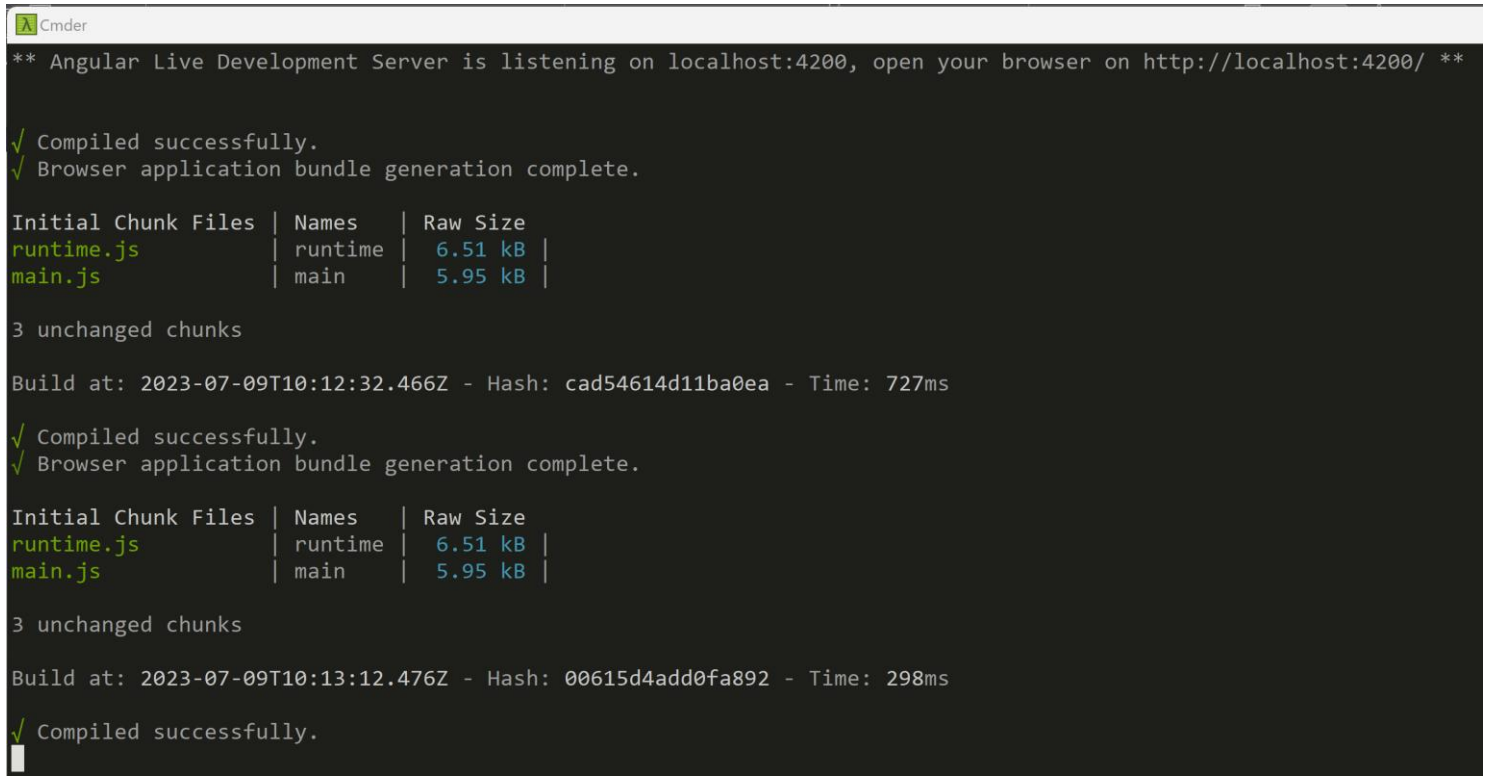


Demo … Live reload

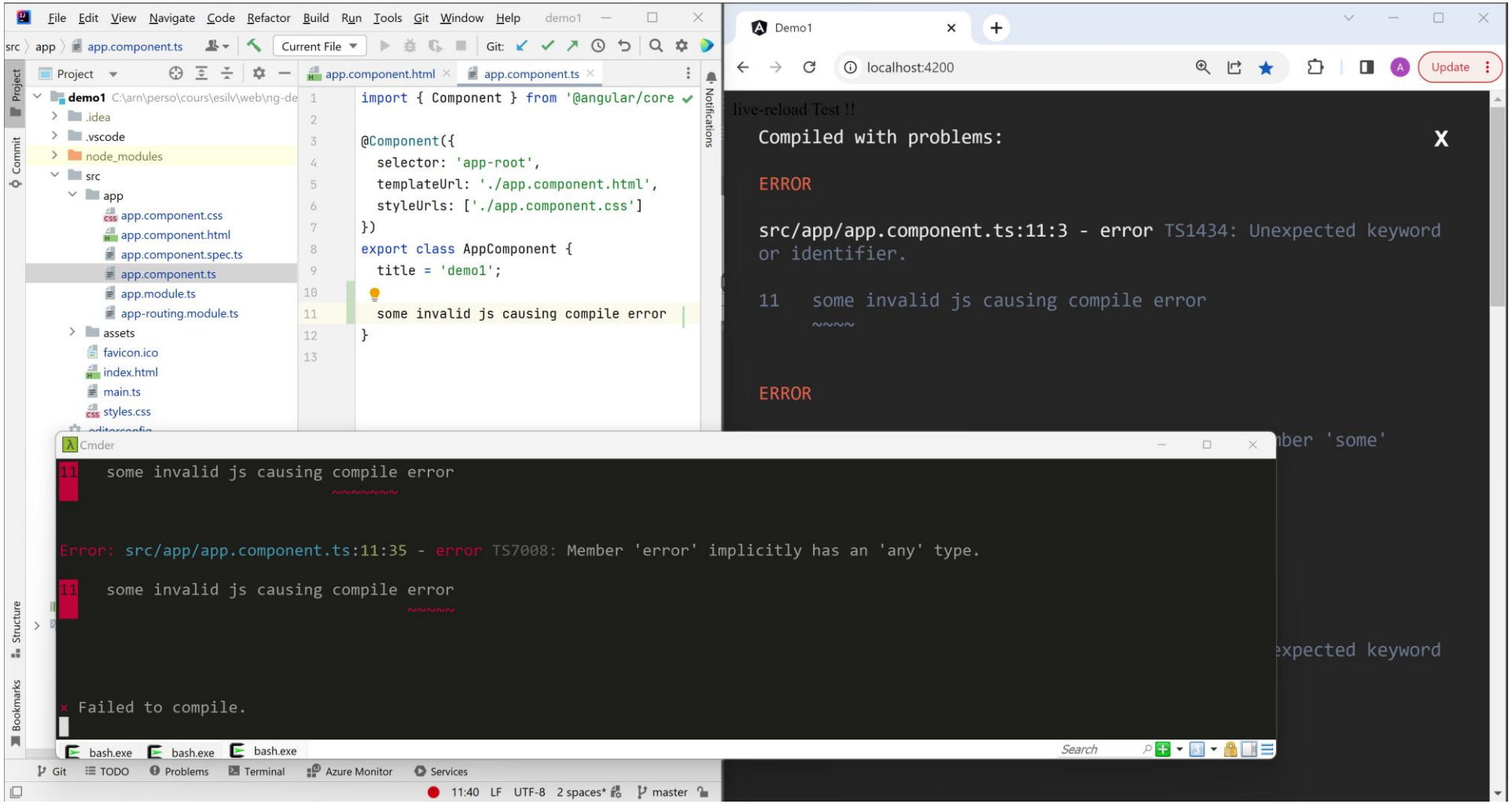


Live Reload … --watch
(expect 5 seconds delay)

watch File changes => compile => refresh page

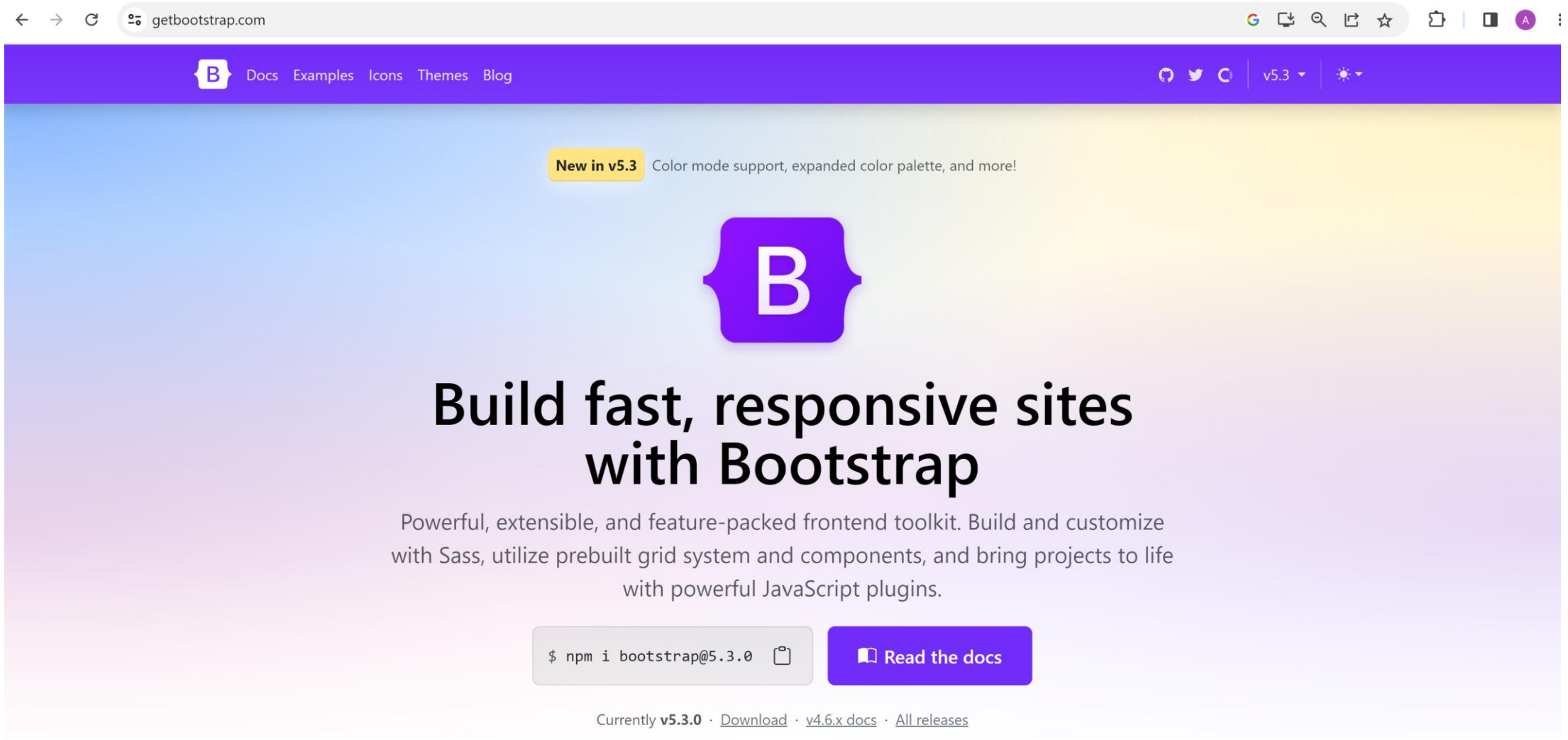


Demo live-reload … when compile error

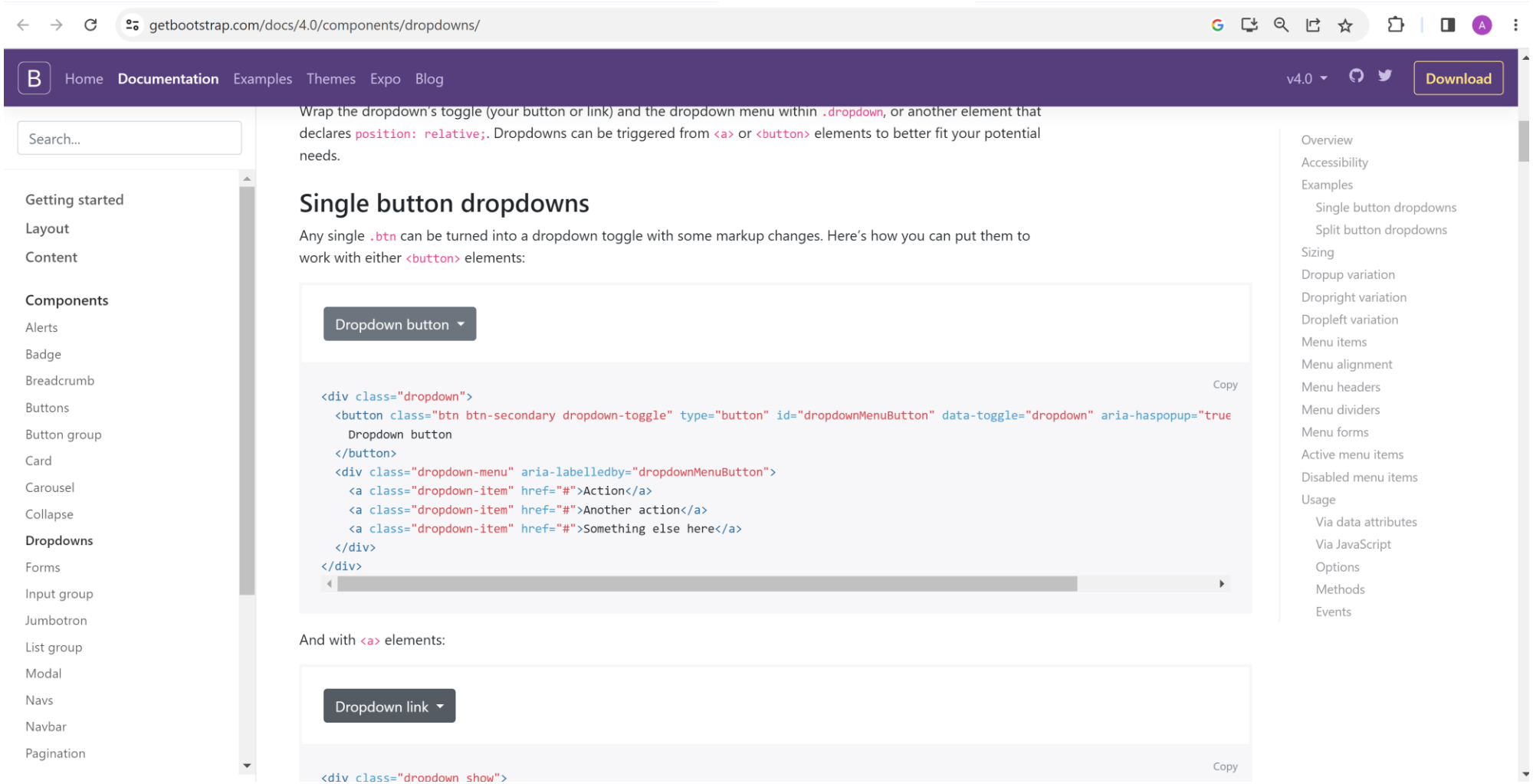


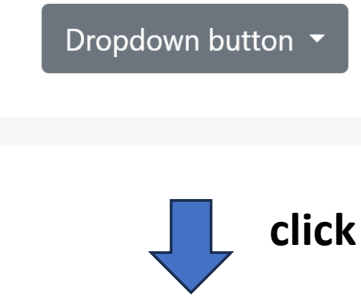
Adding Bootstrap library

https://getbootstrap.com/

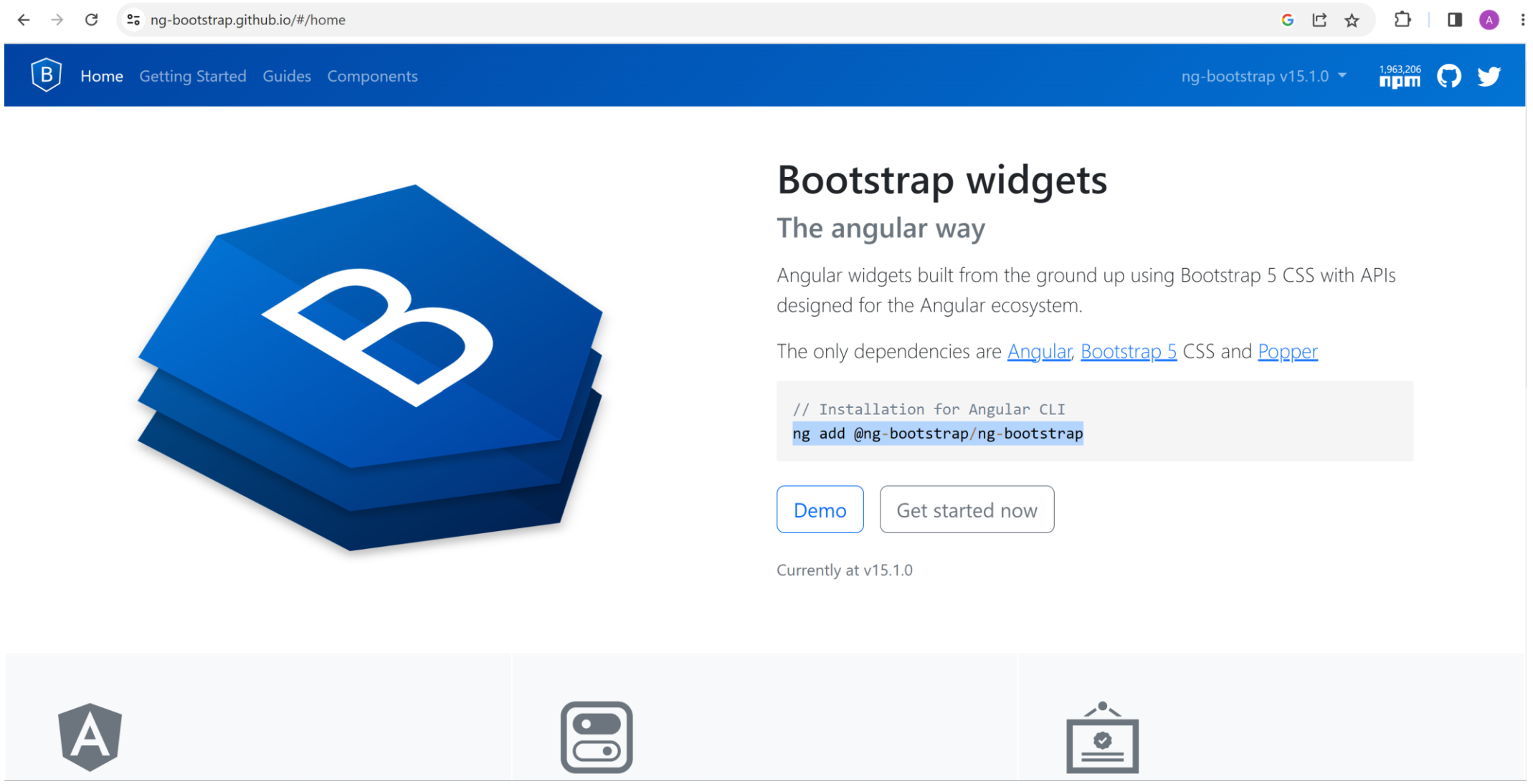


Bootstrap … example: button dropdown





Adding via ng-bootstrap
… NPM module for Bootstrap in Angular



ng add @ng-bootstrap/ng-bootstrap

```
$ ng add @ng-bootstrap/ng-bootstrap
i Using package manager: npm
√ Found compatible package version: @ng-bootstrap/ng-bootstrap@14.2.0.
√ Package information loaded.

The package @ng-bootstrap/ng-bootstrap@14.2.0 will be installed and executed.
Would you like to proceed? Yes
√ Packages successfully installed.
UPDATE package.json (1176 bytes)
√ Packages installed successfully.
Option "name" is deprecated: Use the "project" option instead.
UPDATE src/app/app.module.ts (464 bytes)
UPDATE angular.json (2762 bytes)
UPDATE src/main.ts (259 bytes)
UPDATE tsconfig.app.json (294 bytes)
UPDATE tsconfig.spec.json (300 bytes)
```

git status ; git diff

```
$ git st
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   angular.json
        modified:   package-lock.json
        modified:   package.json
        modified:   src/app/app.module.ts
        modified:   src/main.ts
        modified:   tsconfig.app.json
        modified:   tsconfig.spec.json
```

```
$ git diff
diff --git a/angular.json b/angular.json
index 02ba507..4085657 100755
--- a/angular.json
+++ b/angular.json
@@ -25,6 +25,7 @@
                "src/assets"
              ],
              "styles": [
+               "node_modules/bootstrap/dist/css/bootstrap.min.css",
                "src/styles.css"
              ],
              "scripts": []
diff --git a/package-lock.json b/package-lock.json
index 5e1cf92..f4f0d61 100755
--- a/package-lock.json
+++ b/package-lock.json
@@ -16,6 +16,9 @@
        "@angular/platform-browser": "^15.0.0",
        "@angular/platform-browser-dynamic": "^15.0.0",
        "@angular/router": "^15.0.0",
+       "@ng-bootstrap/ng-bootstrap": "^14.2.0",
+       "@popperjs/core": "^2.11.6",
+       "bootstrap": "^5.2.3",
        "rxjs": "~7.5.0",
        "tslib": "^2.3.0",
        "zone.js": "~0.12.0"
@@ -24,6 +27,7 @@
        "@angular-devkit/build-angular": "^15.0.0",
        "@angular/cli": "~15.0.0",
        "@angular/compiler-cli": "^15.0.0",
+       "@angular/localize": "^15.0.0",
        "@types/jasmine": "~4.3.0",
        "jasmine-core": "~4.5.0",
        "karma": "~6.4.0",
@@ -38,7 +42,6 @@
      "version": "2.2.0",
```

REDO npm install
+ stop & restart ng server

```
$ npm install

removed 1 package, and audited 969 packages in 12s

109 packages are looking for funding
  run `npm fund` for details

22 moderate severity vulnerabilities

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
```

Bootstrap Demos

Bootstrap demo 1:
Container, 12 x columns system

```
<div class="container">
  <div class="row">
    <div class="col-md-1">col</div>
    <div class="col-md-1">col</div>
    <div class="col-md-1">col</div>
    <div class="col-md-3">col md-3</div>
    <div class="col-md-6">col md-6</div>
  </div>
</div>
```

```
  </div>
  <div class="row">
    <div class="col-md-2 app-border1">col md-2</div>
    <div class="col-md-2 app-border1">col md-2</div>
    <div class="col-md-2 app-border1">col md-2</div>
    <div class="col-md-2 app-border1">col md-2</div>
    <div class="col-md-2 app-border1">col md-2</div>
    <div class="col-md-2 app-border1">col md-2</div>
  </div>
  <div class="row">
    <div class="col-md-3 app-border1">col md-3</div>
    <div class="col-md-3 app-border1">col md-3</div>
    <div class="col-md-3 app-border1">col md-3</div>
    <div class="col-md-3 app-border1">col md-3</div>
  </div>
  <div class="row">
    <div class="col-md-4 app-border1">col md-4</div>
    <div class="col-md-4 app-border1">col md-4</div>
    <div class="col-md-4 app-border1">col md-4</div>
  </div>
  <div class="row">
    <div class="col-md-6 app-border1">col md-6</div>
    <div class="col-md-6 app-border1">col md-6</div>
  </div>
</div>
```

Bootstrap demo 1:
Container, 12 x columns system

Test Bootstrap - 12 columns system

Test Bootstrap class=container, class=row, class=col-*

col col col col md-3 col md-6

... More Test Bootstrap class=container, class=row, class=col-*

col md-1	col md-1	col md-1	col md-1	col md-1	col md-1	col md-1	col md-1	col md-1	col md-1	col md-1	col md-1
col md-2		col md-2		col md-2		col md-2		col md-2		col md-2	
col md-3			col md-3			col md-3			col md-3		
col md-4				col md-4				col md-4			
col md-6						col md-6					

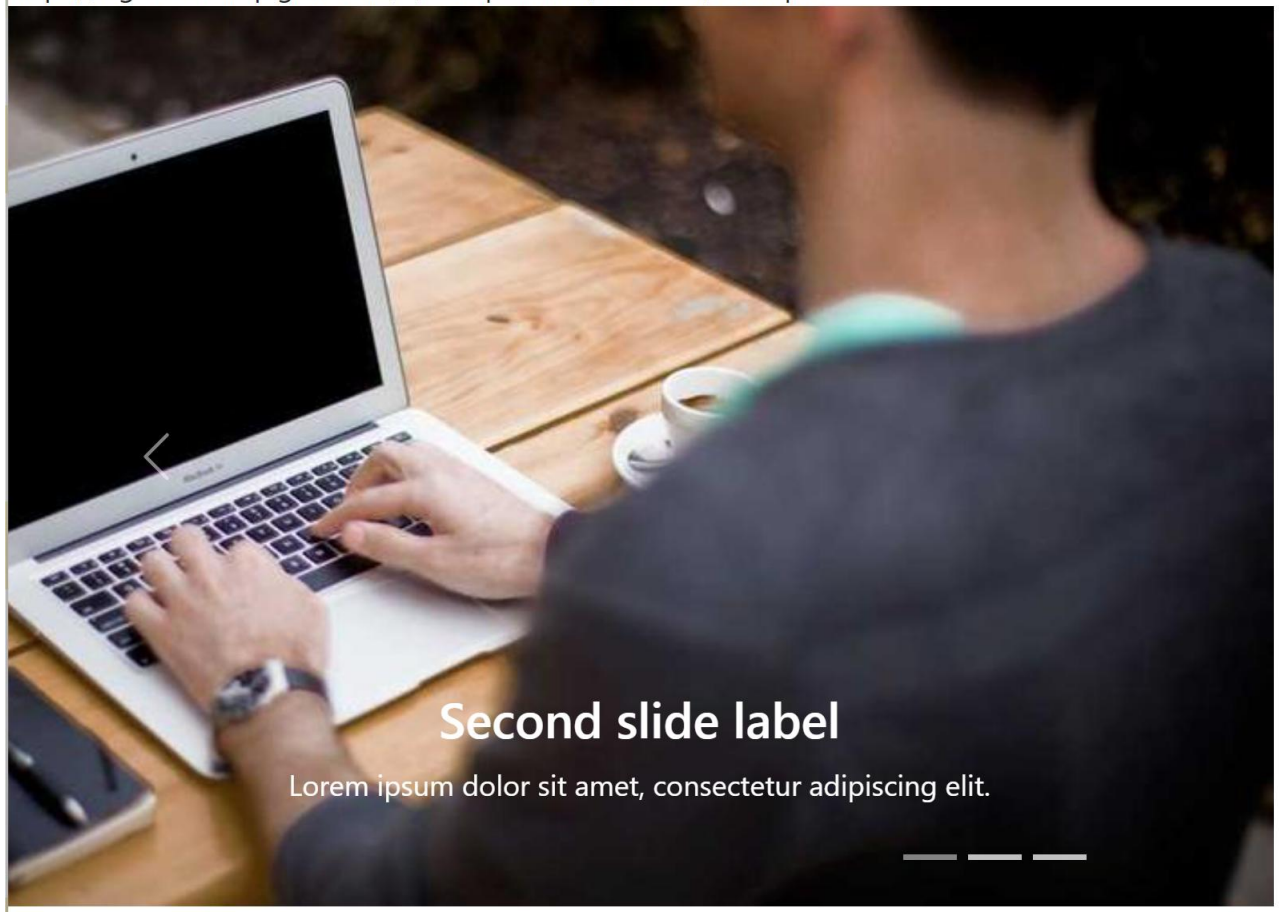
Demo Bootstrap Carousel

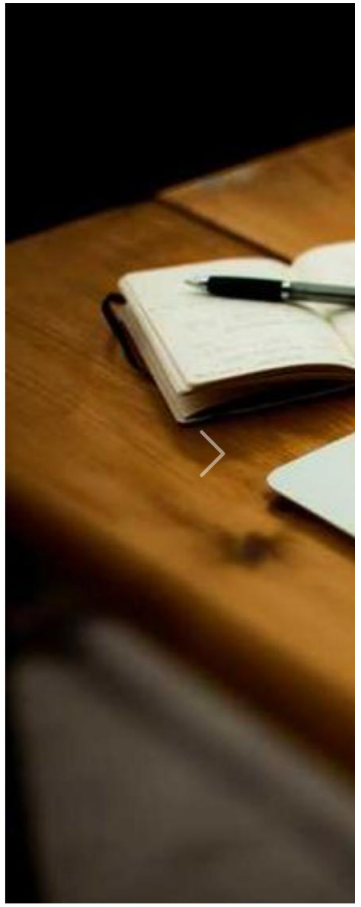
```
<ngb-carousel>
  <ng-template ngbSlide>
    <div class="picsum-img-wrapper">
      <img src="https://picsum.photos/id/0/900/500" alt="Random first slide"/>
    </div>
    <div class="carousel-caption">
      <h3>First slide label</h3>
      <p>Nulla vitae elit libero, a pharetra augue mollis interdum.</p>
    </div>
  </ng-template>
  <ng-template ngbSlide>
    <div class="picsum-img-wrapper">
      <img src="https://picsum.photos/id/1/900/500" alt="Random second slide"/>
    </div>
    <div class="carousel-caption">
      <h3>Second slide label</h3>
      <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit.</p>
    </div>
  </ng-template>
```

Bootstrap demo: Carousel

demo-bootstrap-carousel works!

https://ng-bootstrap.github.io/#/components/carousel/examples





Bootstrap Button

```
<button type="button" class="btn btn-primary">Primary</button>  
<button type="button" class="btn btn-secondary">Secondary</button>  
<button type="button" class="btn btn-success">Success</button>  
<button type="button" class="btn btn-warning">Warning</button>  
<button type="button" class="btn btn-danger">Danger</button>  
```

```
<button type="button" class="btn btn-outline-primary">Primary</button>  
<button type="button" class="btn btn-outline-secondary">Secondary</button>  
<button type="button" class="btn btn-outline-success">Success</button>  
```

```
<button type="button" class="btn btn-primary btn-sm">Small button</button>  
<button type="button" class="btn btn-primary">Std size</button>  
<button type="button" class="btn btn-primary btn-lg">Large button</button>  
```

Bootstrap demo: Button styles

see https://getbootstrap.com/docs/4.0/components/buttons/

button colour: btn-primary, btn-secondary, btn-warning, ..

Primary Secondary Success Warning Danger

button btn-outlined-*

Primary Secondary Success Warning Danger

button size: btn-sm, bt-lg

Small button Std size Large button

Bootstrap demo:
Button dropdown

```
<A href="https://getbootstrap.com/docs/5.0/components/dropdowns/">https://getbootstrap.com/docs

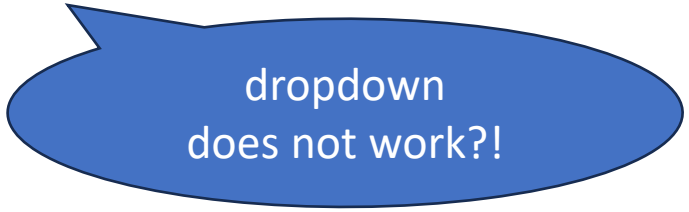
Dropdown button... native class / javascript does not work?!..

<div class="dropdown">
  <button class="btn btn-secondary dropdown-toggle" type="button"
          id="dropdownMenuButton1" data-bs-toggle="dropdown" aria-expanded="false">
    Dropdown button
  </button>
  <ul class="dropdown-menu" aria-labelledby="dropdownMenuButton1">
    <li><a class="dropdown-item" href="#">Action</a></li>
    <li><a class="dropdown-item" href="#">Another action</a></li>
    <li><a class="dropdown-item" href="#">Something else here</a></li>
  </ul>
</div>
```

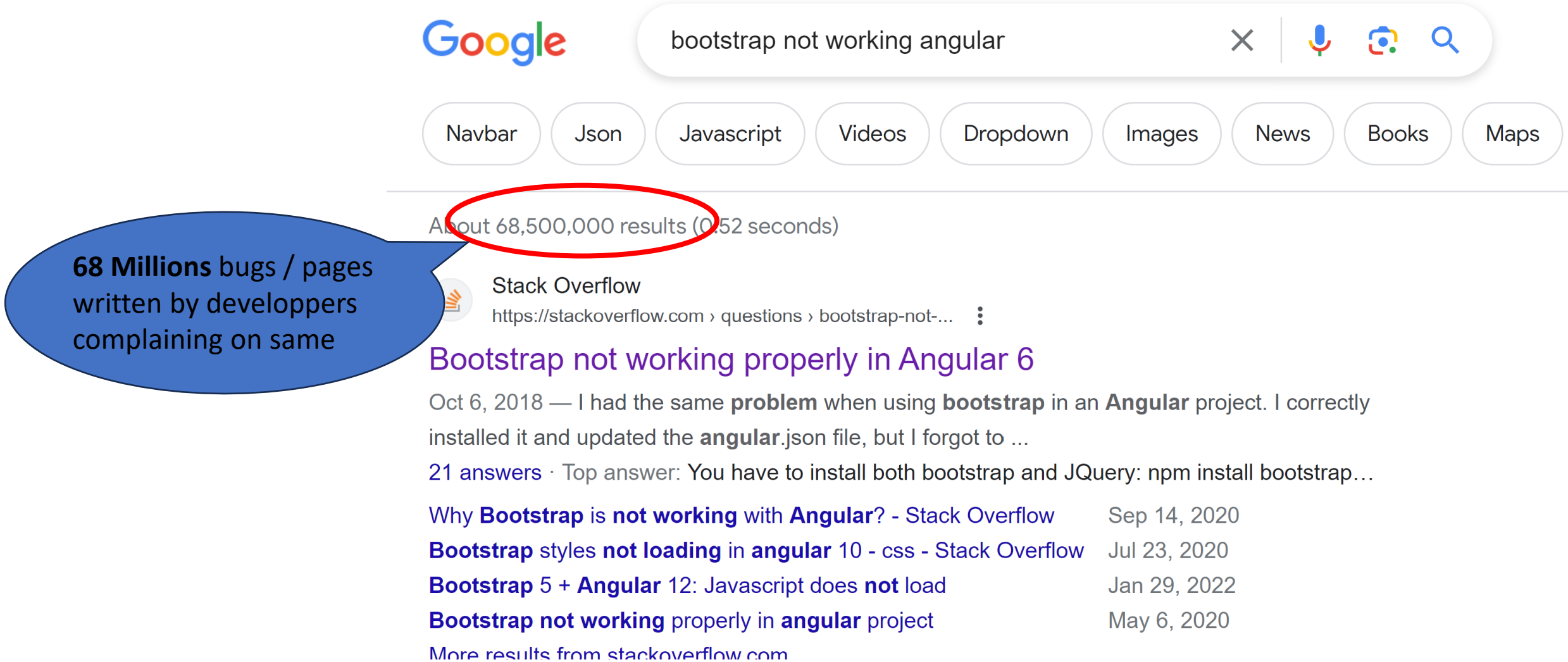
Bootstrap demo:
Button dropdown

https://getbootstrap.com/docs/5.0/components/dropdowns/ Dropdown button... native class / javascript does not work?!..

Dropdown button

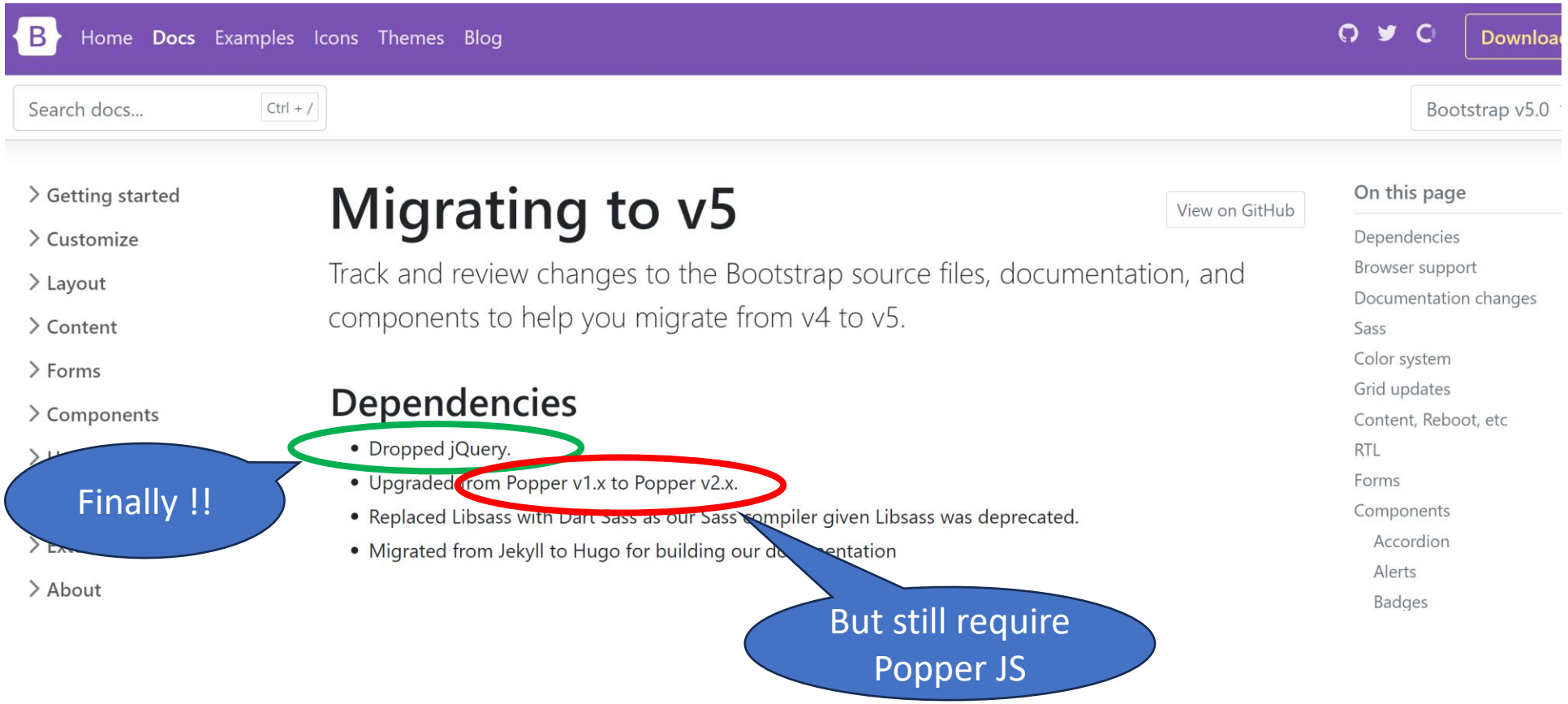


!!! Demo may not work ??
You are not alone

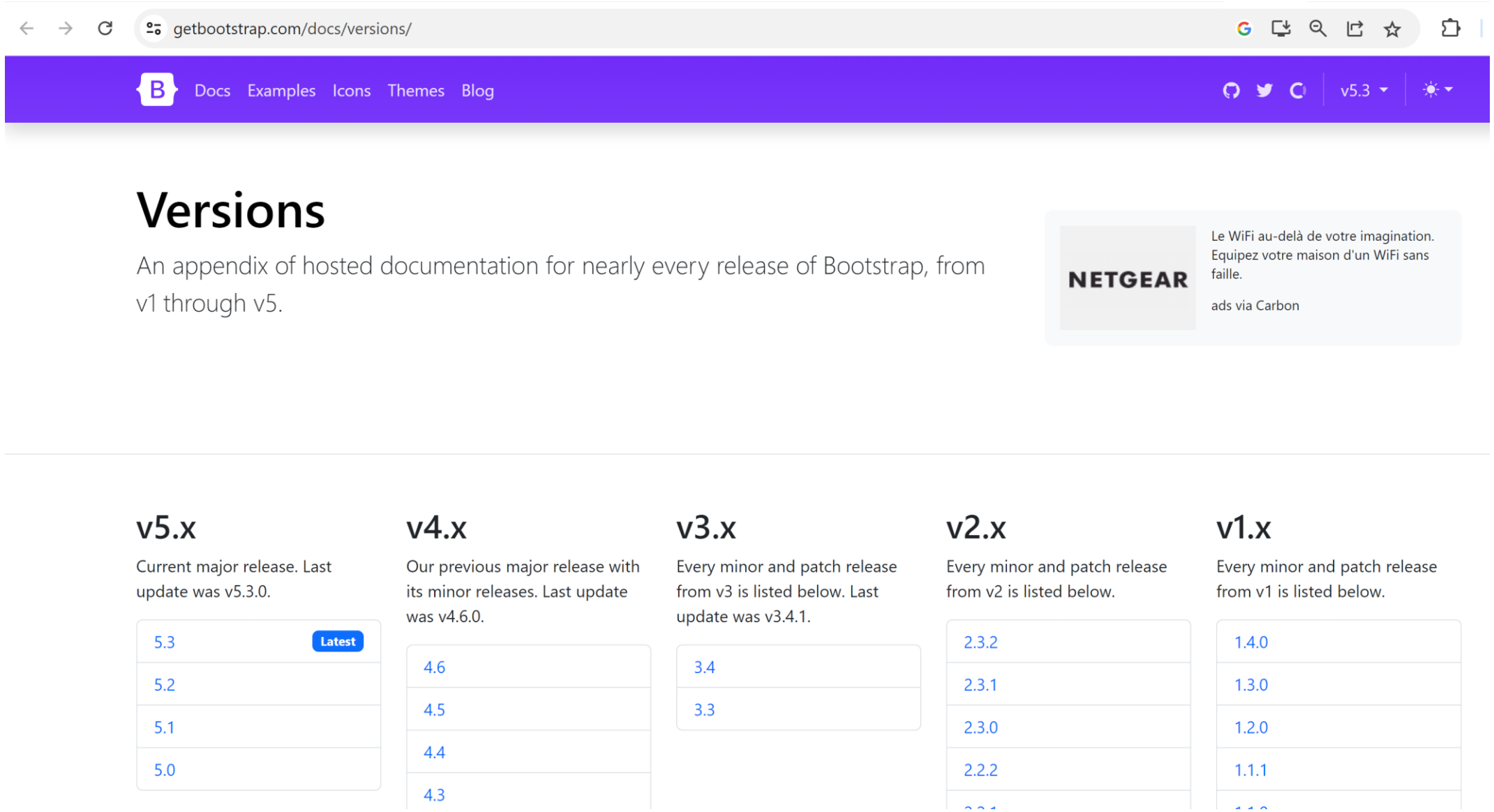


68 Millions bugs / pages written by developpers complaining on same

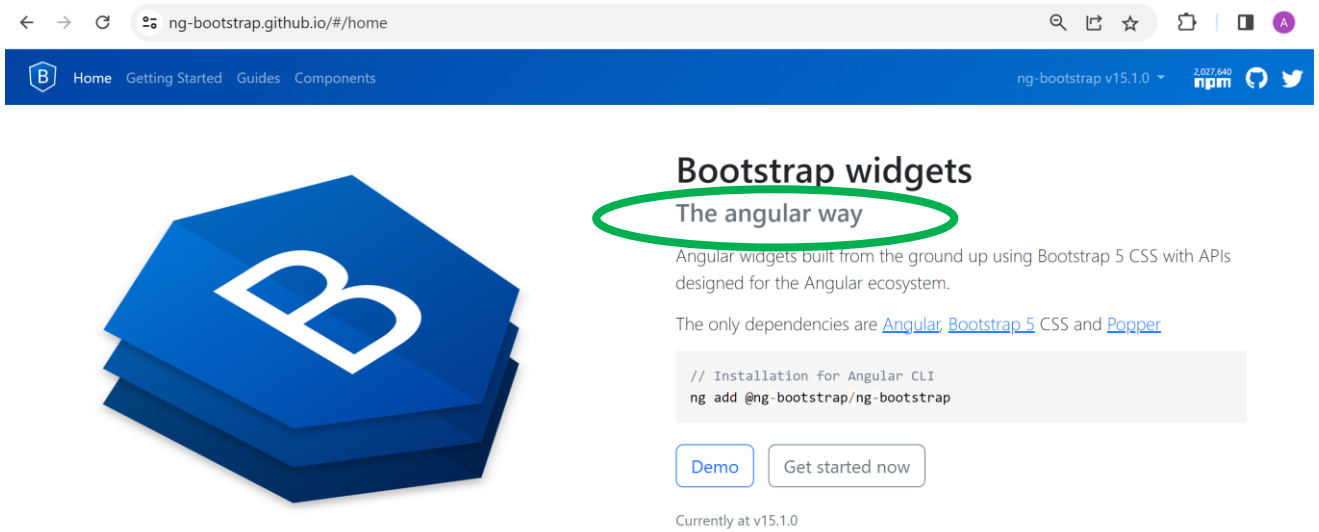
Root cause of all problems: CSS depends on JS



Bootstrap major versions



Testing basic bootstrap with dynamic javascript features => using via « ng-bootsrap »



Some « Vanilla » Bootstrap
Classes + « data-toggle »=.. attributes

Replace by ..

simpler ng-bootstrap equivalent

ngbDropDown
ngbToggle
ngb..

Ngb = initials for
a**ng**ular **B**ootstrap

Bootstrap demo: Button dropdown

```html
<div class="dropdown">
  <button class="btn btn-secondary dropdown-toggle" type="button"
          id="dropdownMenuButton1" data-bs-toggle="dropdown" aria-expa
    Dropdown button
  </button>
  <ul class="dropdown-menu" aria-labelledby="dropdownMenuButton1">
    <li><a class="dropdown-item" href="#">Action</a></li>
    <li><a class="dropdown-item" href="#">Another action</a></li>
    <li><a class="dropdown-item" href="#">Something else here</a></li>
  </ul>
</div>
```



```html
<div ngbDropdown class="d-inline-block">
  <button type="button" class="btn btn-secondary"
          id="dropdownMenuButton2" ngbDropdownToggle>
    Toggle dropdown
  </button>
  <ul ngbDropdownMenu aria-labelledby="dropdownMenuButton2">
    <li><a ngbDropdownItem href="#">Action</a></li>
    <li><a ngbDropdownItem href="#">Another action</a></li>
    <li><a ngbDropdownItem href="#">Something else here</a></li>
  </ul>
</div>
```

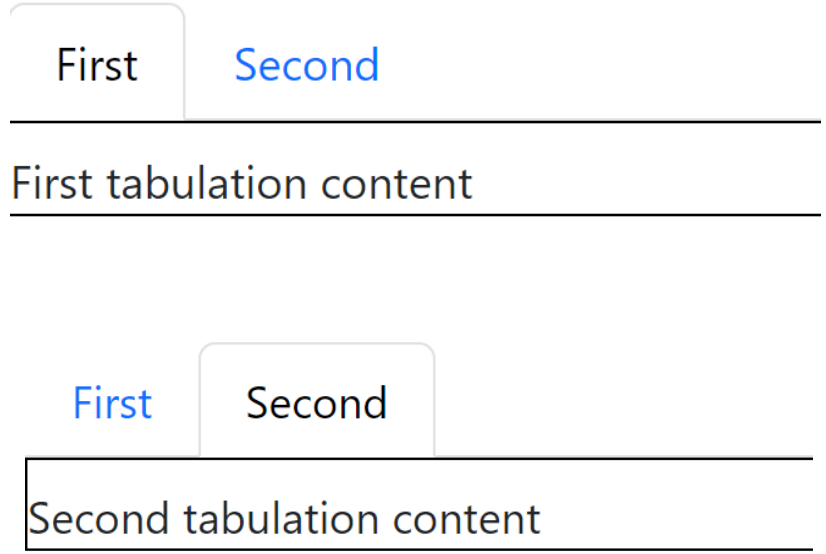
Bootstrap demo:
Button ngbDropdown

data-bs-toggle="dropdown"	-->	ngbDropdown
class="dropdown-toggle"	-->	ngbDropdownToggle
class="dropdown-menu"	-->	ngbDropdownMenu
class="dropdown-item"	-->	ngbDropdownItem



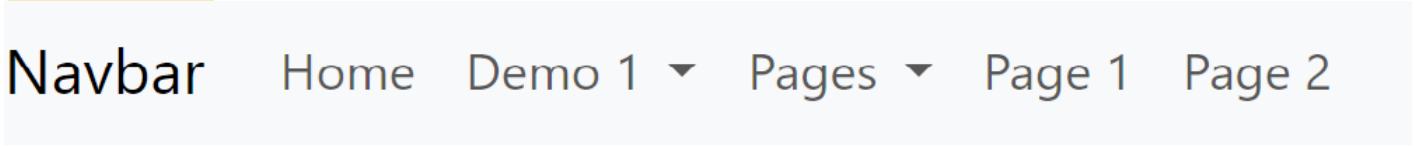
Bootstrap demo: Nav Tabulations

```
<ul ngbNav #nav="ngbNav" class="nav-tabs">
  <li ngbNavItem>
    <button ngbNavLink>First</button>
    <ng-template ngbNavContent>First tabulation content</ng-template>
  </li>
  <li ngbNavItem>
    <button ngbNavLink>Second</button>
    <ng-template ngbNavContent>Second tabulation content</ng-template>
  </li>
</ul>
<div class="app-border1">
  <div [ngbNavOutlet]="nav" class="mt-2"></div>
</div>
```



Bootstrap demo: Navbar Menu

```
<nav class="navbar navbar-expand-lg navbar-light bg-light">
  <a class="navbar-brand" href="#">Navbar</a>
  <ul class="navbar-nav mr-auto">
    <li class="nav-item active">
      <a class="nav-link" href="#">Home</a>
    </li>
    <li ngbDropdown class="nav-item">
      <a ngbDropdownToggle id="navbarMenuDemo1" class="nav-link" role="button">
        Demo 1
      </a>
      <div ngbDropdownMenu aria-labelledby="navbarMenuDemo1" class="dropdown-menu" >
        <a class="dropdown-item" routerLink="/demo1/demo-bootstrap-column">Bootstrap 12 columns</a>
        <a class="dropdown-item" routerLink="/demo1/demo-bootstrap-carousel">Bootstrap Carousel</a>
        <a class="dropdown-item" routerLink="/demo1/demo-bootstrap-button">Bootstrap Button</a>
        <a class="dropdown-item" routerLink="/demo1/demo-bootstrap-button-dropdown">Bootstrap Butto
        <a class="dropdown-item" routerLink="/demo1/demo-bootstrap-navtab">Bootstrap Nav Tabulation
      </div>
    </li>
    <li ngbDropdown class="nav-item">
```



Summary

Install node.js (npm)
$ npm Install –g @angular/cli
$ ng new
=> file: package.json, angular.json
app.component.{html | ts}

$ ng serve
$ ng add @ng-bootstrap/ng-boostrap
$ npm install

<div class=« container »>
<div class=« row »>
<div class=« col-mb-2 »>

<button type=« button » class=« btn btn-primary »>

<nav>

Take Away

$ ng new + ng serve + ng generate + …

=> GET easily a complete Angular environment
… pre-packaging angular + npm + typescript + …

Add (ng-)bootstrap
=> GET basic CSS + elements
… 12-columns layout system
+ button, menu nav bar, tab, etc.

Questions

Next Step

{{interpolation}}
*ngFor
*ngIf
@Component
[] .. @Input (=One way binding)
() .. @Output
[()] .. 2-Ways binding
Router

next document:
https://github.com/Arnaud-Nauwynck/presentations/web/angular-demos-2-ngFor-ngIf-component.pdf